

Muhammad Zafar Ul Haq

AI/ML Projects | NLP & Computer Vision | Teaching Assistant | Tech Community Leadership

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PROFILE

AI/ML-focused project builder with hands-on work in machine learning, NLP, computer vision, semantic search, RAG-based features, and FastAPI applications. Combines applied AI projects with teaching support, research exposure, and student tech-community leadership.

TECHNICAL SKILLS

AI/ML: Scikit-Learn, XGBoost, Logistic Regression, Random Forest, PyTorch, TensorFlow, Keras, CNNs, NLP, Transformers, BERT, GPT-2

LLM/RAG & Data: LangChain, Ollama, Sentence-Transformers, Embeddings, Semantic Search, pgvector, RAG, Pandas, NumPy, Matplotlib

Web/Backend: Next.js, React, TypeScript, FastAPI, REST APIs, PostgreSQL, Redis/Celery, Docker Compose, MySQL, MongoDB

Programming/Tools: Python, C++, JavaScript, SQL, Git, GitHub, Jupyter Notebook, Google Colab, VS Code

EXPERIENCE

Teaching Assistant - OOP & Programming Fundamentals

Oct 2025 - Present

University of Central Punjab

Lahore

- Served as OOP TA from Oct 2025 to Mar 2026; currently assisting Programming Fundamentals labs, evaluations, debugging support, and student problem-solving.

CAP Mentor / Ambassador Program

Mar 2025 - Present

Devsinc

Lahore

- Support the next Devsinc Ambassador Program by mentoring ambassadors, helping with campus coordination, and guiding outreach activities.
- Recognized through Silver, Gold, Platinum Ambassador, and CAP Idol levels for consistent student engagement and community contribution.

AI Intern

May 2025 - Jun 2025

Kangaroo Ventures

Remote

- Completed guided AI projects including recommendation system, NLP resume parser, traffic sign classifier, DistilBERT/GPT-2 chatbot, and sentiment analysis.

Student Pedagogy Partner

Mar 2025 - Feb 2026

Learning Innovation Centre - UCP

Lahore

- Worked with faculty on coding exercises, algorithm simulations, coding challenges, assessment improvement, and learning support material.

Chair / Interim Chair

Jun 2024 - Oct 2025

IEEE Computer Society UCP Student Chapter

Lahore

- Helped establish and lead the chapter; supported workshops, technical sessions, industry collaborations, and student engagement activities.

SELECTED PROJECTS

- Protons EduVerse - Final Year Project** – AI-powered study abroad platform built with Next.js, React, TypeScript, FastAPI, PostgreSQL/pgvector, Redis/Celery, and Docker. Worked on EduVerse AI chat, SOP/LOR generation, admission prediction, semantic scholarship search, and university/faculty/scholarship data pipelines.
- Loan Default Prediction using FT-Transformer** – Compared Logistic Regression, XGBoost, and FT-Transformer on LendingClub data using AUC, F1-score, KS statistic, class-imbalance handling, and threshold tuning.
- Resume Semantic Search** – Built an NLP-based matching system using resume parsing, embeddings, semantic search, ranking, and candidate matching logic.
- Self-Healing Retrieval Assistant** – Created a RAG-style assistant that reads local files, retrieves relevant context, and retries when the first answer is weak or incomplete.
- Additional Mini Projects** – Virtual tutor, face blur, facial expression recognition, activity recognition, object detection, traffic sign classification, sentiment analysis, sales analytics, text-to-SQL, and Chrome extension automation.

RESEARCH

- Loan Default Prediction Using FT-Transformer on LendingClub Financial Data** - Manuscript in Progress; working on mixed tabular feature tokenization, class imbalance, AUC/F1/KS evaluation, and Transformer-based credit-risk modeling.
- Redefining Object Detection: Harnessing the Full Potential of YOLO** - Published, March 2025; focused on YOLO-based object detection, real-time detection concepts, model evaluation, and computer vision applications.

EDUCATION

BS Computer Science, University of Central Punjab

Oct 2022 - Jun 2026

CGPA: 3.8/4.00 | 100% Merit Scholarship Holder

FSc Pre-Engineering, Hadaf College, Peshawar

Apr 2019 - Jun 2021

AWARDS, CERTIFICATIONS & ADDITIONAL

Awards: 100% Merit Scholarship, District Talent Award, OED Supervisor Award, District Level AFAQ Quiz Competition, Hadaf Group of Colleges Recognition.

Leadership: CAP Idol Recognition - Devsinc, Platinum/Gold/Silver Ambassador Recognition, Chair - IEEE Computer Society UCP Student Chapter.

Certifications: Deep Learning Specialization, Machine Learning, Python for Data Science, Introduction to HTML5, Scam Awareness Course.

Languages: Khowar, Urdu, English, Pashto.